

Leveraging R Markdown for Process Narratives

Brendan Dugan, Research Associate, National Survey of Student Engagement



Abstract

R Markdown¹ while often used for data analysis or teaching, can be employed to create detailed narratives for ETL processes, to supplement data dictionaries, and to help analysts share and document their work. This poster describes the process of consolidating SQL queries and SPSS scripts for creating survey weights into an R Markdown document with annotations describing each step in detail, and highlight advantages of this approach.

Background

Recent high churn in higher education^{2, 3} coupled with typically undersized IR/IE shops⁴ can complicate training and knowledge transfer. Departures of long-tenured staff can create gaps in institutional or unit knowledge that can enable mistakes and require additional time learning or reinventing tasks and processes.

Data definitions and analytic decisions are not always formalized, stable, explicit, or well communicated. Infrequent tasks (IPEDS data submissions, annual surveys) or *ad hoc* tasks, complex statistical analyses, and data cleaning pipelines all benefit from detailed *process narratives* describing purpose, definitions or conventions, scope, and methods, above and beyond “commenting your code.”

These narratives also assist in explaining processes to non-technical audiences and code to novice or less-fluent analysts (e.g. R to an SPSS or STATA user or Python programmer) in a way that is simpler than interpreting code or flowcharts.

Example: Creating survey weights

NSSE collects population files from institutions containing student email addresses and variables used in statistical weighting to adjust for known differences in engagement and response rates: enrollment status (full- or part-time), gender (male, female, another e.g. non-binary, unknown), and class level (first-year or senior).

Each student's response to the survey is assigned a disposition code such as “Complete,” “Partial,” or “Explicit refusal,” that determine inclusion in reports and in constructing sample weights⁵. For instance, NSSE includes only students with complete or partial survey responses, and imputes weights for cells with fewer than five respondents.

What is the goal? Describe the general aims of the task.

Why is this done? Describe the history or broader context.

What does it affect? Who should do it? Describe where it fits into broader processes if not self-contained. Include definitions, acronyms, and dataset variables as needed.

What does this task consist of?

Include table previews, plots, as well as flowcharts or other illustrations.

Include code with plain-language descriptions. Lengthier pipes or sections can be broken into smaller unevaluated chunks (*r, echo=TRUE, eval=FALSE*) as examples preceding the hidden evaluated section.

Identify distinct task components and organize by headers (H2, H3, etc.)

What other resources would help?

Process Narrative Example

Brendan Dugan
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Purpose

To construct statistical weights for use in reporting NSSE survey responses to institutions.

Background

As part of each survey administration, NSSE collects population files from institutions containing student email addresses and variables used in statistical weighting to adjust for known differences in engagement and response rates: enrollment status (full- or part-time), gender (male, female, another e.g. non-binary, unknown), and class level (first-year or senior). Each student's response to the survey is assigned a disposition code (American Association for Public Opinion Research, 2023) such as “Complete,” “Partial,” “Explicit refusal,” or “Logged on to the survey, did not complete any items” that determine inclusion in reports and in constructing sample weights.

Scope

Lead: Brendan Dugan
Support: ...

As they feature weights or weighted statistics, the tables processed here affect final dataset delivery and report creation.

Timeline

Generally this occurs in late May or early June following the closing of the survey and the Report Form.

Definitions

- **population**: All first-year and senior bachelor's degree-seeking students, regardless of age, residence, or enrollment status
- **sample**: All census- or randomly-sampled students, often the entire sample (~9.5% are census sample)
- **respondents**: All sample students with complete or partial dispositions
- **disposition**: Type of response to survey based on AAPOR framework

Process

Data are extracted from the database table `population`, which features response disposition and student demographic information. Weights are computed by class, gender, and enrollment status from their respective representation in the population and respondents.

Extract Data

Extract, apply common filters, further subset to sample and respondents, then combine in one table.

unitid	class	enrollment	gender	pop	sample	resp
123456	fy	Full-time	Male	102	102	46
123456	fy	Full-time	Female	277	277	51
123456	fy	Part-time	Male	19	19	1
123456	fy	Part-time	Female	43	43	14
123456	sr	Full-time	Male	107	107	35
123456	sr	Full-time	Female	303	303	89
123456	sr	Part-time	Male	15	15	4
123456	sr	Part-time	Female	32	32	12

Compute Weights and Response Rates

Weights adjust `resp` counts to reflect their number in the `population` and to account for different known response patterns. Weights are given as $w_i = P_i / r_i$. The mean of `weight_scale` is used to adjust `weight_adj`.

`adj(w_i) = w_i / mean(w_i)` if `i` is not empty (0), it is treated as missing; between 1 and 4, it's the overall weight; otherwise, just as calculated.

```
# the main output
weights <- group_by(dispositions, unitid, class) |>
  mutate(
    sample_temp = if_else(resp < 5, NA, sample),
    resp_temp = if_else(resp < 5, NA, resp),
    weight_scale_overall = sum(sample_temp) / sum(resp_temp),
    # only calculate weights for cells with at least 5
    weight_scale = case_when(resp >= 5 ~ sample / resp,
                             resp < 5 & resp > 0 ~ weight_scale_overall,
                             resp < 5 & NA ~ NA_real_),
    weight_adj = weight_scale / weight_scale_overall,
    rr = resp / sample * 100,
    sample_temp = NULL,
    resp_temp = NULL,
    weight_scale_overall = NULL
  )
```

unitid	class	enrollment	gender	pop	sample	resp	weight_scale	weight_adj	rr
123456	fy	Full-time	Unknown	28	28	9	3.22	0.14	30.77
123456	fy	Full-time	Male	102	102	46	3.96	0.90	25.27
123456	fy	Full-time	Female	277	277	51	5.43	1.23	10.41
123456	fy	Full-time	Another sex	24	24	6	4.00	0.91	25.00
123456	fy	Part-time	Unknown	4	4	0	NA	NA	0.00
123456	fy	Part-time	Male	19	19	1	4.42	1.00	5.26
123456	fy	Part-time	Female	43	43	14	3.07	0.70	32.36
123456	fy	Part-time	Another sex	1	1	0	NA	NA	0.00

Label and Export Data

Variable labels are added for export in SPSS and coupled with data in an R-native format. This could similarly be maintained within a database environment or Excel workbook.

Resources

- The American Association for Public Opinion Research. 2023 *Standard Definitions: Final Dispositions of Case Codes and Outcome Rates for Surveys*. 10th edition. AAPOR. Retrieved from <https://aapor.org/wp-content/uploads/2024/03/Standard-Definitions-10th-edition.pdf>
- National Survey of Student Engagement. “An Explanation of Weighting in the NSSE Institutional Report.” <https://nsse.indiana.edu/news/reports/data/weighting.html>

This had been accomplished by a series of SQL queries to filter and count these groups at different levels of aggregation (population, sample, respondents) then exported and modified in SPSS to merge with other data locally.

```
--sample counts
TRANSFORM Count(population.unitid) AS countofunitid
SELECT unitid
FROM population
WHERE (class="First-year" Or class="Senior")
AND sample IN ("Census", "Random sample", "Random oversample")
AND dispoCode IN (1,1, 1,2, 2,111, 2,121, 2,12, 3,19)
AND (enrollment="Full-time" Or enrollment="Part-time")
AND gender IN ("Male", "Female", "Unknown", "Another sex")
GROUP BY unitid
PIVOT class;
```

However, these queries would be repeated and modified to account for different filter criteria, and required multiple software and sequential extracts. This process was rewritten in R and mapped to avoid repeating code, and blended with clear explanations to avoid confusion.

```
# sample: map(population, filter, ...) |> map(group_by) |> ...
filter(population,
  (class == "First-year" | class == "Senior") &
  sample %in% c("Census", "Random sample", "Random oversample") &
  dispoLabel %in% c("Complete", "Partial", "Explicit refusal",
    "Logged on to survey, did not complete any items",
    "Break-off", "Nothing returned") &
  (enrollment == "Full-time" | enrollment == "Part-time") &
  gender %in% c("Male", "Female", "Unknown", "Another sex")) |>
  group_by(unitid) |>
  count(class)
```

Conclusion

Though it requires greater initial investment and foresight, creating process narratives can save time in training, contextualizing, and revisiting tasks, make it easier to collaborate, and encourage good documentation.



Template and example:
github.com/bjdugan/AIR24

References

1. Allaire, J., Xie, Y., Dervieux, C., McPherson, J., Luraschi, J., Ushey, K., Atkins, A., Wickham, H., Cheng, J., Chang, W., Iannone, R. (2023). *markdown: Dynamic Documents for R*. R package version 2.25. <https://github.com/rstudio/markdown>
2. Fried, M. (2023). *The Great Resignation and Higher Education Employees*. Ithaka S+R. <https://sr.ithaka.org/blog/the-great-resignation-and-higher-education-employees/>
3. Lederman, D. (2022). *Turnover, Burnout and Demoralization in Higher Ed*. Inside Higher Ed. <https://www.insidehighered.com/news/2022/05/04/turnover-burnout-and-demoralization-higher-ed>
4. Jones, D., Keller, C., & Raza, Z. (2022). *2021 AIR National Survey of IR Offices: IR Office Staff Roles [Report]*. Association for Institutional Research. www.airweb.org/NationalSurvey
5. The American Association for Public Opinion Research. (2023). *2023 Standard Definitions: Final Dispositions of Case Codes and Outcome Rates for Surveys*. 10th edition. AAPOR. Retrieved from <https://aapor.org/wp-content/uploads/2024/03/Standard-Definitions-10th-edition.pdf>